

The visual phototransduction pathway in mammals is the process by which visual stimulation in the form of photons of light is biochemically received and converted into electrical signals that can be processed by the brain to form images. This phototransduction pathway begins when photons strike rod and cone photoreceptor cells located at the back of the retina. Instead of dendrites, each photoreceptor nerve cell contains hundreds of lipid membrane discs, each of which is filled with opsins, a type of G protein–coupled receptor.

In place of a conventional agonist, opsins are instead covalently bound to a prosthetic group known as retinal. Retinal is attached to the opsin protein through the formation of a Schiff base between the electrophilic aldehyde group of retinal and a nucleophilic primary amine of an amino acid side chain. In the dark, the retinal prosthetic group exists in the 11-*cis* conformation, but energy input from photons causes isomerization to the all-*trans* conformation. In the presence of the light-stimulated all-*trans* isomer, the opsins activate their associated heterotrimeric G protein, known as transducin.

Once activated, transducin then activates cGMP phosphodiesterase, which catalyzes the degradation of 3',5'-cyclic GMP (cGMP). Cyclic nucleotide–gated (CNG) channels are Na⁺- and Ca²⁺-permeable channels that open upon binding to cGMP, allowing extracellular cations to enter. With the degradation of cGMP, these CNG channels close, causing gradual changes in the photoreceptor membrane potential and gradual changes in neurotransmitter release. In the retina, the inputs from the various rod and cone cells are processed by various interneurons—bipolar cells, horizontal cells, and amacrine cells—each with their own graded membrane potential, before reaching the ganglion cells at the front side of the retina. The ganglion cells then process all excitatory and inhibitory inputs in an all-or-none fashion to either produce or not produce an action potential that then travels to the brain for further cognitive processing.

Based on the passage, which of the following amino acid side chains is most likely involved in the formation of the Schiff base that links to retinal?

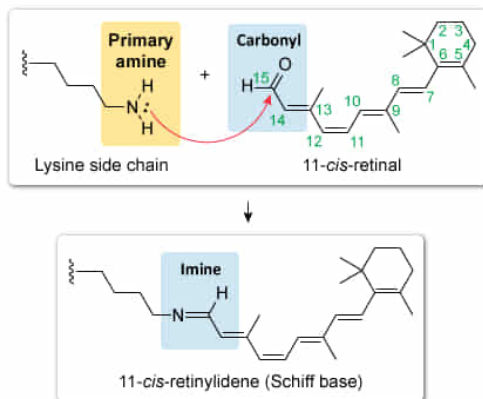
- ☐ A. Serine [8%]
- ✓ ☒ B. Lysine [64%]
- ☐ C. Histidine [6%]
- ☐ D. Asparagine [20%]

Correct

64% answered correctly

Explanation:

A lysine side chain reacts with retinal to form a Schiff base



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In biochemistry, Schiff bases appear when a protein forms a covalent bond to a carbonyl-containing compound by the nucleophilic **addition of an amine** to form an **imine** linkage. Schiff bases are extremely important in biochemistry, serving as reaction intermediates in enzyme-catalyzed reactions or as covalent linkages to **prosthetic groups**, as described in the passage.

The passage states that Schiff base formation involves the aldehyde group of retinal being attacked by a **nucleophilic primary amine** from an **amino acid side chain**. Of the amino acids listed, only **lysine** contains a primary amine that can react with 11-*cis* retinal to form a Schiff base.

(Choice A) **Serine** contains a hydroxyl group in its side chain. The side chain does *not* contain a primary amine.

(Choice C) The nitrogen atoms in the side chain of **histidine** are each bonded to more than one carbon; therefore, they are *not* primary amines.

(Choice D) The -NH_2 group in the side chain of **asparagine** exists as part of the larger **amide** functional group. Due to **resonance** with the adjacent carbonyl group, the lone pair of electrons on the nitrogen is unlikely to act as a nucleophile.

Educational objective:

Schiff bases are commonly seen in biochemistry as imine linkages between a protein and a substrate or a prosthetic group. Schiff bases are formed between a primary amine (eg, of a lysine side chain) and a carbonyl compound.